Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/12/2007
PI Name:	Sivathanu, Yudaya R. Ph.D.		
Project Title:	A Time-Resolved Micro-Impedance Tomography System for the Determination of Capillary Flow Instability Phenomenon		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sivathan@enurga.com	**Fax:**	FY 765-463-7004
PI Organization Type:	INDUSTRY	**Phone:**	765-497-3269
Organization Name:	En'Urga Inc.		
PI Address 1:	Suite A102		
PI Address 2:	1291 Cumberland Avenue		
PI Web Page:	http://www.enurga.com		
City:	West Lafayette	**State:**	IN
Zip Code:	47906	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	05/01/2005	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 (from 9/30/2006) per PI (11/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feikema , Douglas (NASA GRC) Mi, Ye (En'Urga Inc.) Meyer, William (NASA GRC)		
Grant/Contract No.:	NNC05GA57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A thorough understanding of two-phase flow, fluid stability and dynamics is very important in a number of manned and unmanned space missions as addressed at the Two-Phase Flow, Fluid Stability and Dynamics Workshop held at NASA Glenn Research Center (GRC) in May 2003. Issues considered “critical” are the primary two-phase instability mechanisms that could be substantially different in space than on Earth. In particular, two-phase flow instabilities are encountered in cryogenic fluid systems, phase separators, thermal management systems, bioreactors, and PEM fuel cells. Two-phase capillary flows exhibit a wide range of behavior, ranging from fully dispersed to annular flow, in normal and reduced gravity. In many instances, the structure of this two-phase flow is poorly understood during transitory flow and through porous medium due to the difficulties in performing reliable measurements at the microscale. The proposed four year ground study consisted of two key elements that will improve the understanding of two-phase flow instabilities. The first element is the design and development of a rugged micro-impedance tomography system for obtaining flow characteristics such as interfacial area, film thickness, and void fractions in two-phase capillary flows with sufficiently high temporal and spatial resolution. The second element is the study of two-phase capillary flow under normal and reduced gravity environments to elucidate the physical phenomena involved in capillary flow instabilities. The two innovative aspects of the proposed instrumentation were: (1) a flexible array of micro-impedance sensors that can be wrapped around small capillary pipes of circular or non-circular cross-section to obtain path integrated impedances at multiple locations enabling instability phenomenon to be elucidated at the micro-scale, and (2) an innovative maximum likelihood estimation algorithm in conjunction with a neural network that will provide structure information for capillary two phase flow instabilities. Electrical impedance tomography systems have been developed for non-intrusively monitoring and controlling multiphase flow (with conducting and non-conducting fluids) in a variety of industrial applications. Compared to other tomography systems, electrical impedance tomography is fast, inexpensive, suitable for small pipes or large vessels, and does not involve safety restraints. In addition, current advances in neural networks and deconvolution algorithms enable rapid classification and quantification of a variety of processes. These two innovations were to be utilized in developing the micro-impedance tomography system to study the structure of two-phase flow instabilities in capillary tubes of circular and non-circular cross-sections. The project was funded for only the first two years. Therefore, the scope of the project was changed. During the two year project, we designed and fabricated a Micro-Impedance Tomography System (MITS) equipped with a sensor for one-inch pipe, instead of a sensor for a micro-channel due to the shortened project period. The MITS was built to handle both one- and two-plane measurements. The former can provide two-dimensional phase distributions, while the latter can provide interfacial velocities. The algorithms for phase reconstruction and velocity measurements were developed and programmed in the MATLAB environment. The software required for data acquisition, calibration and a graphic user interface of the MITS were also developed. The whole system was tested in a two-phase flow loop available at En’Urga. Feasibility of the system was demonstrated during the two year project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Two-phase flow instabilities are encountered in cryogenic fluid systems, phase separators, thermal management systems, bioreactors, and PEM fuel cells. The results of this study showed the feasibility of developing an instrument that can be used to better understand and control fluid instabilities in two-phase systems.
Task Progress:	The primary objective of the research was to develop well-validated diagnostics to characterize two-phase capillary flow in normal and microgravity environments. The proposed work was completed with the following two specific tasks: (1) hardware development of a single/double-plane micro-impedance tomography system (MITS), (2) development of reconstruction algorithm. During the project, we designed and fabricated a Micro-Impedance Tomography System (MITS) equipped with a sensor for one-inch pipe, instead of a sensor for a micro-channel due to the shortened project period. The MITS was built to handle both one- and two-plane measurements. The former can provide two-dimensional phase distributions, while the latter can provide interfacial velocities. The algorithms for phase reconstruction and velocity measurements were developed and programmed in the MATLAB environment. The software required for data acquisition, calibration and a graphic user interface of the MITS were also developed. The whole system was tested in a two-phase flow loop available at En’Urga. The feasibility of the system was completely demonstrated in the first two years.
Bibliography Type:	Description: (Last Updated: 01/07/2020)